Ritesh Roshan
Independent Researcher
riteshroshansahoo01@gmail.com
11 November 2026

The Editor-in-Chief
Pattern Recognition

Dear Editor-in-Chief,

I am pleased to submit the manuscript "CFFM-Net: Reliability-Gated Mamba Fusion of Visible and Thermal Images for Detection, Tracking and Multi-Camera Analytics" for consideration as a regular paper in Pattern Recognition. The work concerns object detection and tracking from two image modalities, which I believe falls within the scope of Pattern Recognition.

The manuscript is a controlled study of visible-thermal object detection when a camera misbehaves. CFFM-Net is a dual-stream YOLO26 detector whose fusion block estimates a per-pixel reliability for each camera and uses it to scale the step of a four-direction selective state-space scan. I compare it with a matched concatenation baseline at nano scale, over three seeds on M3FD and LLVIP, and report the results as they came out. Fusing the two cameras beats the best single camera on M3FD by 4.0 mAP, but a plain concatenation baseline captures most of that gain. The gated scan adds a small margin of 0.6 mAP on M3FD that is not significant at three seeds, and nothing measurable on LLVIP. On an eleven-fault benchmark the model leads on thermal-side faults and trails on visible-side ones, and a baseline trained with fault augmentation is more robust than the model. A Temporal State Memory lowers the KAIST miss rate under thermal noise from 51.6 to 34.1 percent, with a confound that I state, and a live runtime keeps delay and GPU memory bounded over ten minutes.

I do not claim state-of-the-art accuracy: the nano-scale models sit below published detectors, and the paper says where each claim stops. I hope that a carefully bounded result of this kind is useful to readers who deploy paired cameras.

The manuscript is original, has not been published, and is not under consideration elsewhere. I am its sole author, and I have no conflicts of interest to declare. The code, configurations and result files are available at https://github.com/riteshroshann/yolo-cross-feature-fusion-mamba-net, and the datasets (LLVIP, M3FD, KAIST and WILDTRACK) are public.

Thank you for considering this work. I would be grateful for the reviewers' comments.

Yours sincerely,

Ritesh Roshan
Independent Researcher
riteshroshansahoo01@gmail.com